The Anatomy of a Class - The Name, Comments, Attributes, methods, Constructors, Accessors, Modeling Real World Systems, Designing with Reuse, Extensibility, Maintainability in Mind and Using Object Persistence;

Programming lessons - Introduction to Java - Java Virtual Machine (JVM) and Java Runtime (JRE), Java Development Kit (JDK), Integrated Development Environment (IDE) for Java, Writing programs in Java and learning Java syntax, features and libraries.

Recommended Texts:

1. M. Weisfeld, *The Object-Oriented Thought Process*, 5th ed. Addison-Wesley Professional, 2019
2. H. Schildt, *Teach Yourself C++*, 2nd ed. McGraw-Hill Osborne Media, 1994

Hum 4242 **Arabic II** **Credit: 1.00**

Reading Comprehension: Use of determiners and pronouns; Use of interrogatives; Use of nominal and verbal sentences Use of adverbs; Use of tenses; Use of Feminine & Masculine Genders; Conjunctive Adverbs; Nouns; Singular; Plural and various modifications caused by them; Use of verbs with different persons and all pronouns; Use of new words (nouns & verbs) by changing different parts of speech.

Hum 4244 **English II** **Credit: 1.00**

This course aims to develop more advanced competencies in international students of English language in reading, writing and comprehending more complex sentence structures, grammatical forms and cohesion. It will lay emphasis on awareness of better precision and fluency of structure, forms, and style. It will teach organization of paragraph, noting salient points, summarizing, writing advanced discourse, reports and stories on familiar and unfamiliar subjects. It will also teach different forms of writing letters, telegrams and applications, besides reporting speeches in indirect forms. It will involve advanced listening and speaking, role-playing, interpreting, discussing, interviewing etc.

CSE 4206 **Digital Logic Design Lab** **Credit: 0.75**

Sessional works based on CSE 4205.

SWE 4202 **Object Oriented Concepts I Lab** **Credit: 1.50**

Sessional works based on SWE 4201.

Third Semester

Math 4341 **Linear Algebra** **Credit: 3.00**

Linear Algebra: Solving $Ax = B$ for square systems by elimination (pivots, multipliers, back substitution, invertibility of A, and factorization into $A = LU$. Complete solution to $Ax = B$ (column space containing b, rank of A, nullspace of A and special solutions to $Ax = 0$ from row reduction).

Basis and dimension (bases for the four fundamental subspaces). Least squares solutions (closest line by understanding projections). Orthogonalization by Gram-Schmidt (factorization into $A = QR$).

Properties of determinants (leading to the cofactor formula and the sum over all $n!$ permutations, applications to inverse matrix calculation and volume). Eigenvalues and eigenvectors (diagonalizing A, computing powers A^k and matrix exponentials to solve difference and differential equations). Symmetric matrices and positive definite matrices (real eigenvalues and orthogonal eigenvectors, tests for $x, Ax > 0$, applications).

Linear transformations and change of basis (connected to the Singular Value Decomposition - orthonormal bases that diagonalize A). Linear algebra in engineering (graphs and networks, Markov matrices, Fourier matrix, Fast Fourier Transform, linear programming).

Recommended Texts:

1. G. Strang, *Introduction to Linear Algebra*, 6th ed. Wellesley-Cambridge Press, 2023

CSE 4303 **Data Structures** **Credit: 3.00**

Introduction to data structures: what & why, Notations, Concept of efficiency. Elementary Data Structures: Arrays, Records & Pointers, Examples of Random Access, Call by Reference, Variable Length Strings, Secondary Storage, and Implementation in Memory. Lists: Concept of Linked Lists.

Lists: The implementation, Sub list, Recursive lists, Variants, Orthogonal lists, Stack & Queue, Sequential & circular implementation of stack & queue, Applications of stack & queue.